ETHAN HANOLD

hanold@wisc.edu | Madison, WI

Education

University of Wisconsin–Madison

PhD in Mathematics — *September 2025 – Present*

Co-advisors: Prof. Qin Li and Prof. Misha Khodak

University of Wisconsin–Madison

Bachelor of Science in Computer Science, Mathematics — *September 2022 - May 2025*

Research

1. **Ethan Hanold**, Nicholas Boffi, Qin Li, Leonardo Zepeda-Núñez. *AdjointFlow: Accelerating Scientific Inverse Problems with Physics-Based Transport.* (in preparation, 2026).

2. Sixu Li, **Ethan Hanold**, Nicholas Boffi, Leonardo Zepeda-Núñez, Qin Li. *When Does Noise Help in Stochastic Interpolants: A Non-Asymptotic Analysis and Optimal Design.* (in preparation, 2026).

Talks & Presentations

1. AdjointFlow: Accelerating Scientific Inverse Problems with Physics-Based Transport. *August 2026*
   SciML for Plasma Workshop, UW–Madison, Madison, WI.

2. AdjointFlow: Accelerating Scientific Inverse Problems with Physics-Based Transport. *March 2026*
   SIAM Conference on Uncertainty Quantification (UQ26), Minneapolis, MN.

3. Stochastic Interpolants for Inverse Scattering. *October 2025*
   3rd RTG NASC Annual Workshop, Rice University, Houston, TX.

4. Machine Learning to Solve Partial Differential Equations. *December 2024*
   Madison Experimental Mathematics Lab, UW–Madison, Madison, WI. Poster Presentation.

Experience

Research Intern, Space Science & Engineering Center (SSEC) — *May - August 2025*
PIs: David Tobin & Willem Marais

Undergraduate Research Assistant, UW-Madison Math Department — *February - May 2025*
PI: Prof. Qin Li

Undergraduate Researcher, Madison Experimental Mathematics Lab (MXM) — *September - December 2024*
Supervisor: Dr. Yukun Yue

Software Development Intern, Epic Systems — *May - August 2024*

Honors & Awards

Linnaeus Wayland Dowling Scholarship — *April 2025*

UW-Madison Dean's List — *Fall 2022-2024, Spring 2023-2025*

William F. Vilas Merit Scholarship — *September 2022*

Teaching

Teaching Assistant: Survey of Calculus 1, UW-Madison *September 2026 - Present*

Teaching Assistant: Survey of Calculus 2, UW-Madison *January - May 2026*

Teaching Assistant: Calculus and Analytic Geometry 1, UW-Madison *September - December 2025*